

Weather Prediction Using Linear Regression

Project Overview

This project focuses on analyzing a weather dataset and building a machine learning model to predict weather-related values. The dataset contains historical weather observations from Seattle, including precipitation, maximum temperature, minimum temperature, wind speed, and weather conditions. The primary objective is to understand the data through Exploratory Data Analysis (EDA) and develop a predictive model using Linear Regression.

Dataset Description

The dataset includes the following features:

- **date:** The date of the weather observation.
- **precipitation:** Amount of rainfall recorded.
- **temp_max:** Maximum temperature of the day.
- **temp_min:** Minimum temperature of the day.
- **wind:** Wind speed measurement.
- **weather:** Weather condition such as sun, rain, drizzle, fog, or snow.

For this project, Linear Regression is used to predict the numerical variable **temp_max** based on other weather-related features.

Methodology

1. Load the dataset using the Pandas library.
2. Perform data preprocessing by checking data types, missing values, and duplicates.
3. Conduct Exploratory Data Analysis (EDA) using summary statistics and visualizations such as histograms, boxplots, and correlation analysis.
4. Select relevant input features (`precipitation` , `temp_min` , and `wind`) and define `temp_max` as the target variable.
5. Split the dataset into training and testing sets.
6. Train a Linear Regression model using Scikit-learn.
7. Generate predictions on the test data.
8. Evaluate the model using metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² Score.
9. Visualize the results with Actual vs. Predicted and Residual plots.

Tools and Libraries

- Python
- Pandas
- NumPy
- Matplotlib

- Scikit-learn

Results

The trained Linear Regression model estimates maximum daily temperature based on selected weather features. The evaluation metrics help measure prediction accuracy, while graphical visualizations provide insight into model performance and error distribution.

Conclusion

This project demonstrates a complete machine learning workflow, from data loading and exploration to model training and evaluation. It highlights the importance of preprocessing, feature selection, and performance assessment. While Linear Regression is suitable for predicting continuous values such as maximum temperature, classification algorithms would be more appropriate if the goal is to predict categorical weather conditions like rain or sunshine.